Project - High Level Design
On
Project Title
AUTONOMOUS RETAIL RESEARCHER

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Introduction

The **Autonomous Retail Researcher Agent** is an advanced AI-driven system designed to autonomously perform retail market research by leveraging LLM-powered agents. The system processes user queries, navigates web sources, extracts relevant information, and generates structured insights for decision-making.
This project aims to automate retail research tasks, reduce manual effort, and provide high-quality, data-driven insights using multi-agent collaboration and intelligent reasoning.

1.1 Scope of the Document

This document provides a high-level architectural overview of the Autonomous Retail Researcher Agent system. It covers system design, workflows, components, APIs, data handling, and non-functional requirements.

1.2 Intended Audience

This document is intended for:
• Software developers and engineers
• System architects
• Project stakeholders
• QA and testing teams

1.3 System Overview

The system consists of multiple intelligent agents working collaboratively:

• **Planner Agent** – Understands user query and creates research plan
• **Web Research Agent** – Searches and collects data from web sources
• **Analysis Agent** – Filters and extracts meaningful insights
• **Writer Agent** – Generates final summarized report

The system integrates APIs, web tools, and LLM models to generate accurate and structured retail insights.

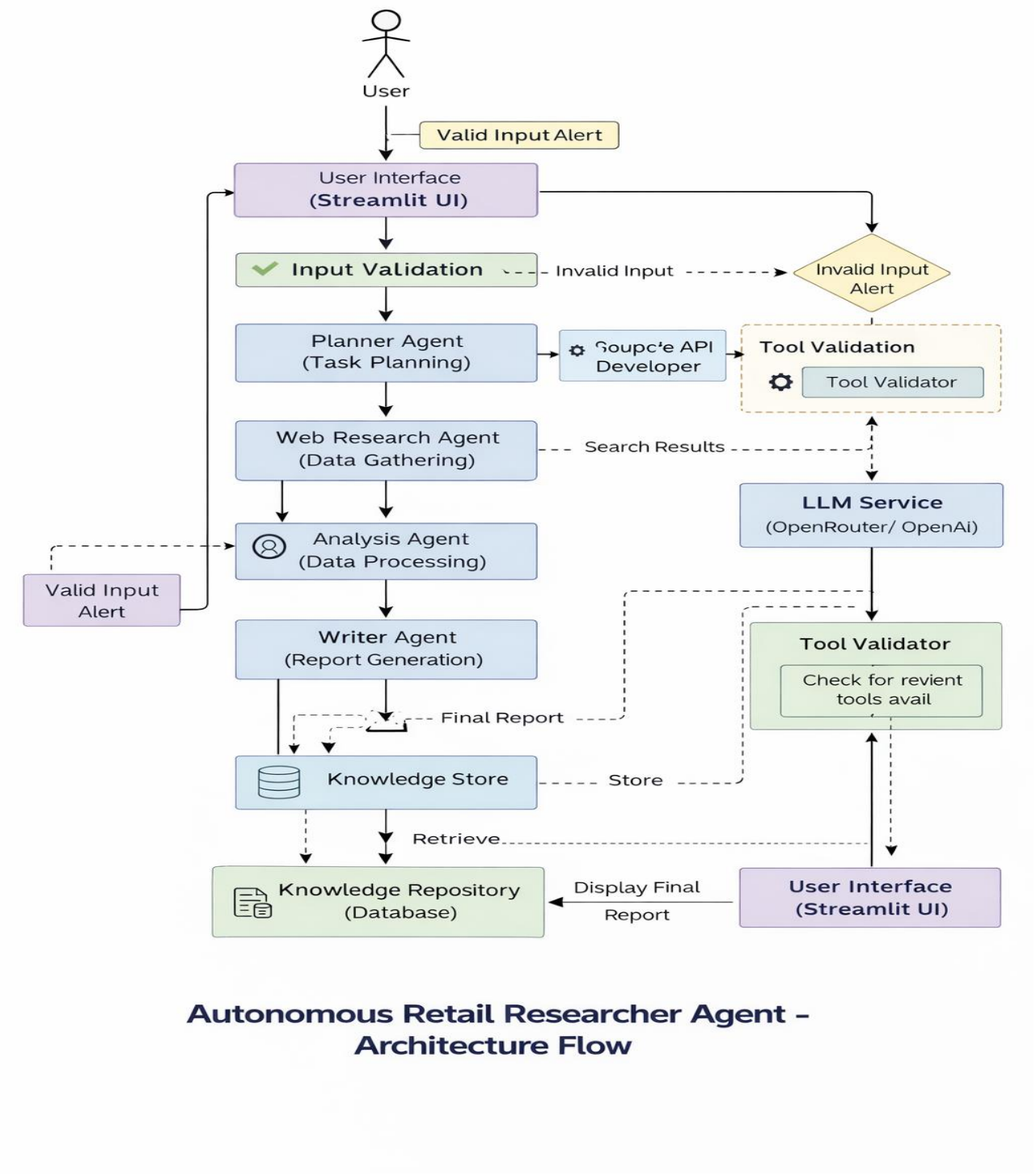
2. System Design

The system follows a modular, scalable, and layered architecture ensuring flexibility and maintainability.

2.1 Application Design

The application is designed using a **multi-tier architecture**:

- **Frontend Layer** – Streamlit UI for user interaction
- **Backend Layer** – Handles request processing and orchestration
- **AI/Agent Layer** – CrewAI + LangChain agents for reasoning and execution
- **Data Layer** – Stores research outputs and knowledge repository

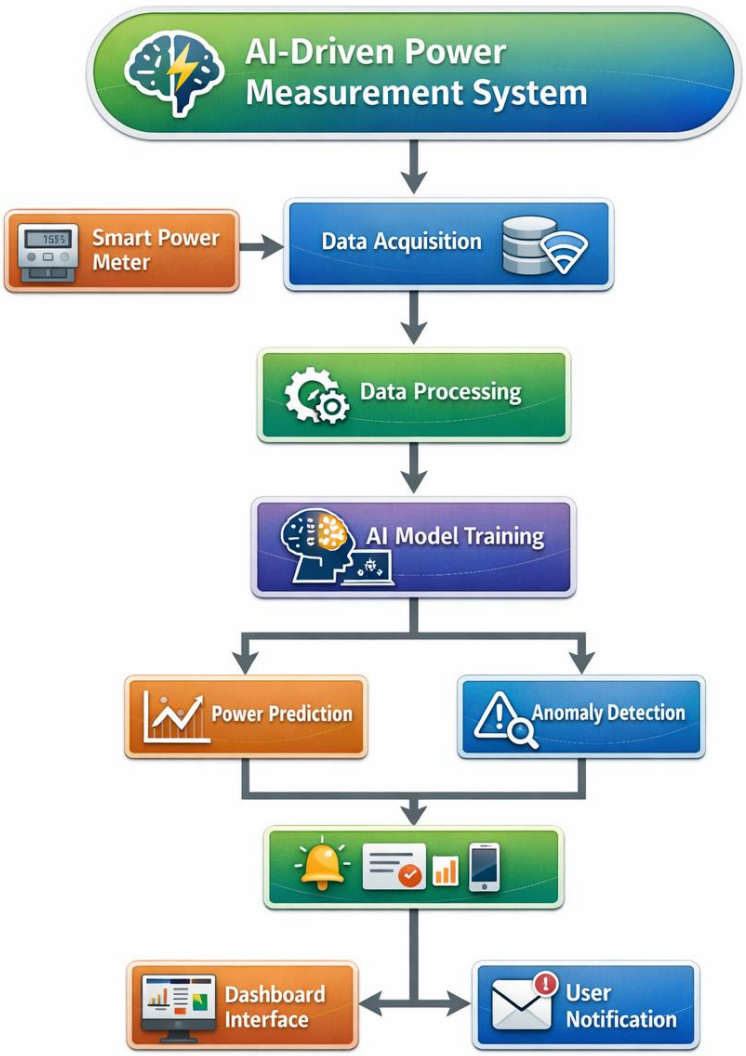


Autonomous Retail Researcher Agent – Architecture Flow

2.2 Process Flow

The process begins when a user enters a retail research query.

1. User submits query via UI
2. Backend receives and validates input
3. Planner Agent decomposes task
4. Web Research Agent gathers data
5. Analysis Agent processes data
6. Writer Agent generates final report
7. Output displayed and stored



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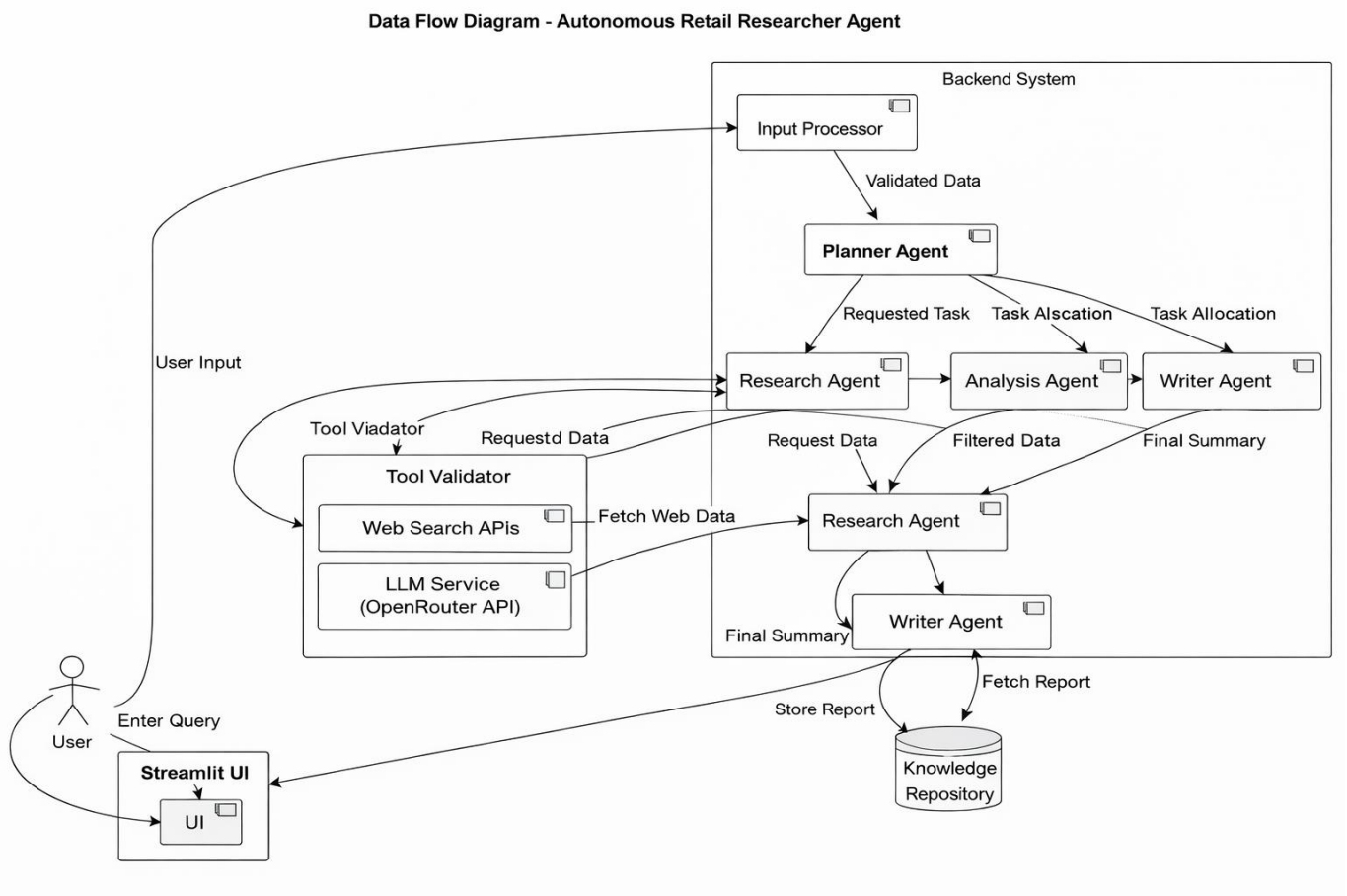
2.3 Information Flow

Data flows through the system as:

- User Input → API Layer → Backend Processing
- Backend → AI Agents → Data Extraction & Analysis
- AI Agents → LLM → Response Generation
- Final Output → UI + Knowledge Repository

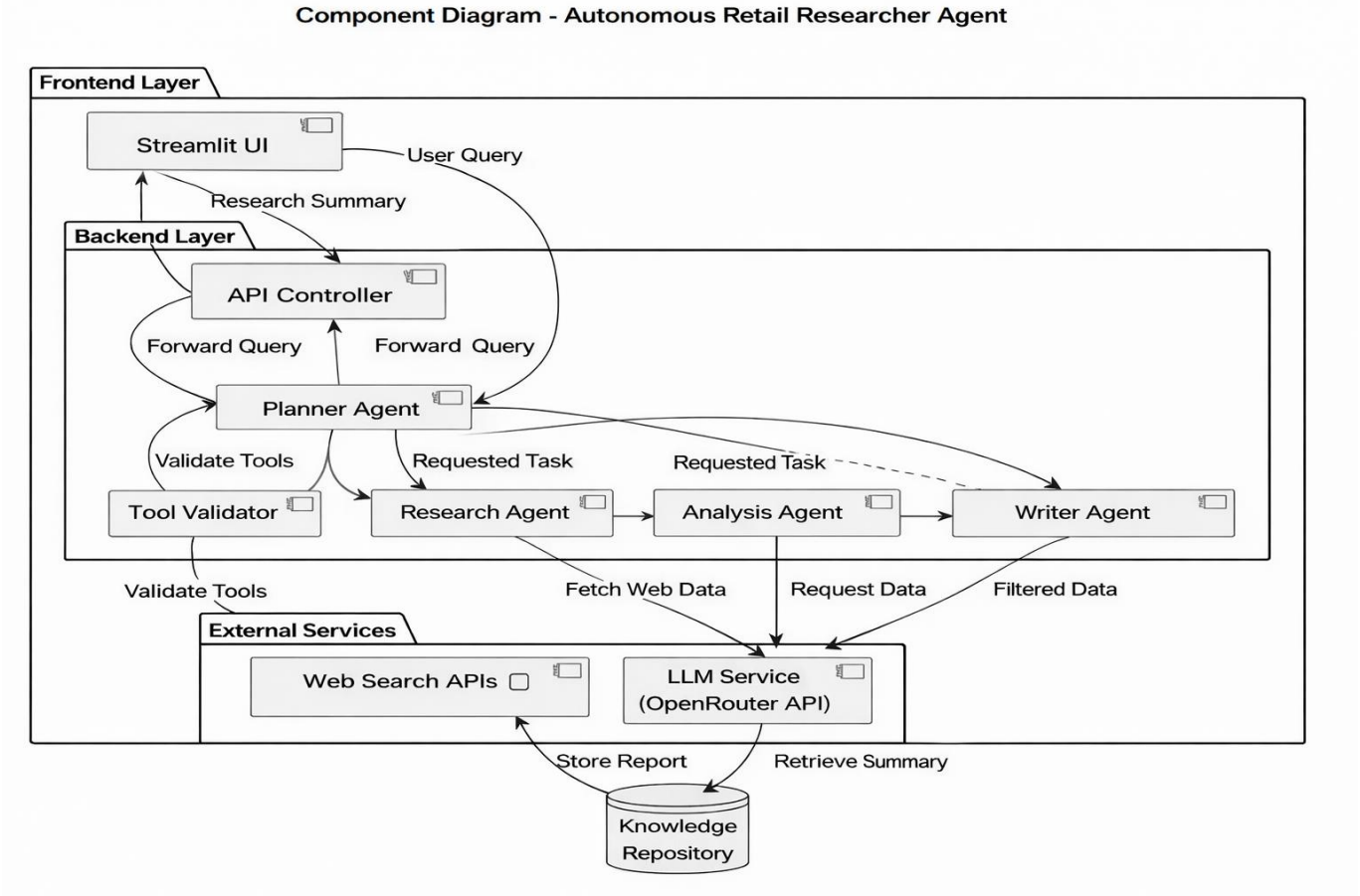
AI agents → Response generation → Frontend display
This ensures smooth communication between components.



2.4 Components Design

The system consists of:

- **User Interface Module** – Handles user interaction
- **API Gateway** – Manages request routing
- **Agent Manager (CrewAI)** – Coordinates agents
- **Web Search Module** – Fetches external data
- **Data Processing Module** – Cleans and analyzes data
- **LLM Service Module** – Generates insights
- **Database System** – Stores outputs and history



2.5 Key Design Considerations

- **Scalability** – Handle large queries and data
- **Reliability** – Ensure consistent output
- **Extensibility** – Add new agents/tools easily
- **Maintainability** – Modular architecture
- **Performance Optimization** – Efficient data handling

2.6 API Catalogue

- **User API** – Accepts user queries
- **Search API** – Fetches web results
- **Analysis API** – Processes extracted data
- **LLM API** – Generates summaries
- **Report API** – Stores and retrieves reports

These APIs follow REST principles for easy integration.

3. Data Design

3.1 Data Model

- **User Entity** – Stores user queries
- **Research Query Entity** – Input queries
- **Source Data Entity** – Collected web data

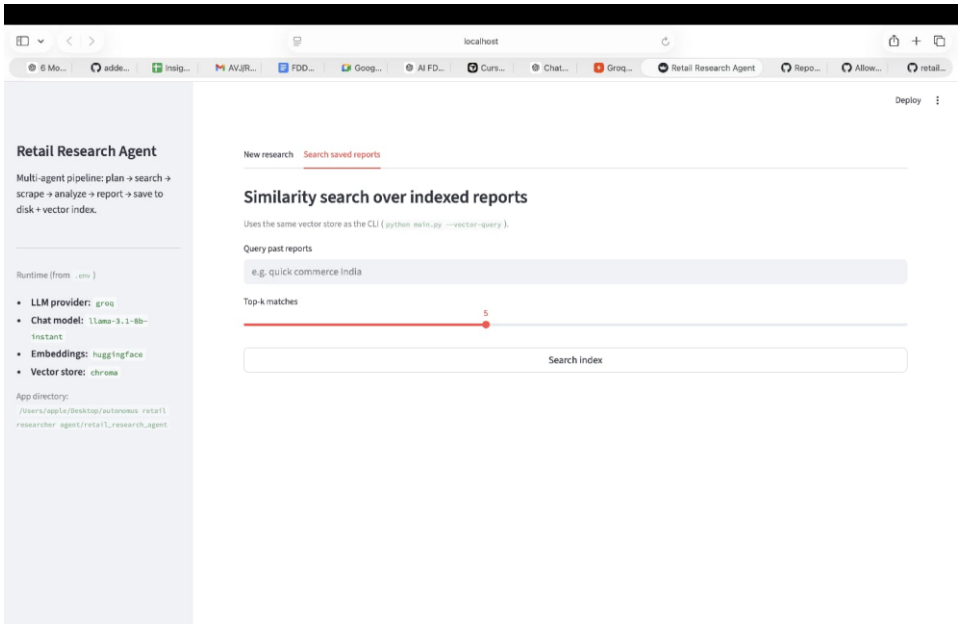
- **Insights Entity** – Processed information
- **Reports Entity** – Final generated reports

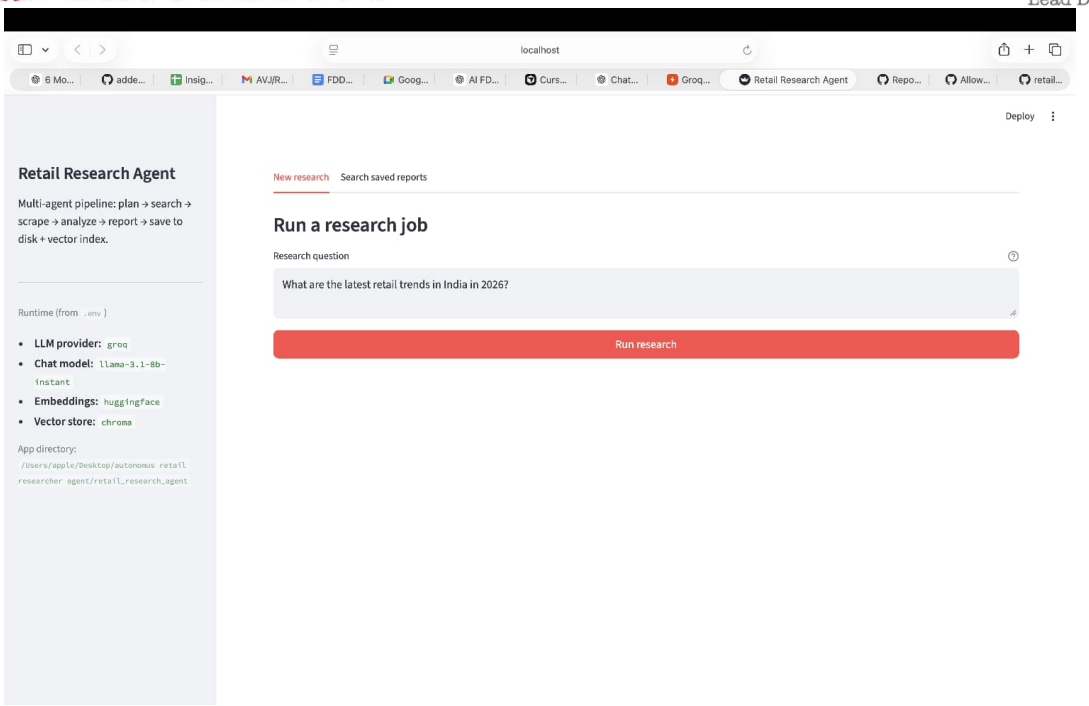
3.2 Data Access Mechanism
- ORM for database interaction
- APIs for external data
- Optimized queries for fast retrieval

3.3 Data Retention Policies
- Temporary data removed after processing
- Research data stored for reuse
- Sensitive data secured and encrypted

3.4 Data Migration
- Backup and restore mechanisms
- Version-controlled updates
- Data transformation tools

4. Interfaces
- **User Interface (UI)** – Streamlit-based frontend
- **API Interfaces** – For integration with services
- **Admin Interface** – Monitoring and management





5. State and Session Management
• Token-based authentication (JWT)
• Session storage for temporary data
• Stateless backend design

6. Caching
• Cache frequently accessed data
• Reduce API and DB load
• Improve response time
• Redis can be used

7. Non-Functional Requirements
7.1 Security Aspects
• Authentication & authorization
• Data encryption
• Secure APIs (HTTPS)
• Protection against attacks

7.2 Performance Aspects
• Efficient algorithms
• Load balancing
• Caching mechanisms
• Scalable infrastructure

8. References

- AI agent system design principles
- LangChain & CrewAI documentation
- REST API design guidelines
- Database design practices